

ORIGINAL ARTICLE

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In-situ parameterization and scaling analysis of CO₂ convective dissolution for offshore saline aquifer sequestration in the South China Sea

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Abstract

Offshore saline aquifers in the South China Sea are promising targets for large-scale CO₂ sequestration because of their substantial capacity and proximity to major coastal emission sources. However, reliable prediction of CO₂ plume migration and dissolution trapping remains challenging because key fluid properties are often estimated from empirical correlations rather than measured under in-situ conditions. This study develops an integrated experimental–numerical framework to investigate the mechanisms and scaling laws of CO₂ convective dissolution. The diffusion coefficient, solubility, and density of the CO₂–brine system were determined from formation brine experiments under reservoir conditions representative of the South China Sea. CO₂ diffusion coefficients measured by Raman spectroscopy range from 2.89×10^{-9} – 4.78×10^{-9} m²·s^{−1} at 313.15–363.15 K and 7.5–17 MPa. Diffusivity increases significantly with temperature, decreases systematically with salinity, and is only weakly affected by pressure. Solubility and density measurements further confirm that CO₂ dissolution generates sufficient density contrast to trigger density-driven convection. These experimentally constrained fluid properties were incorporated into Darcy-scale numerical simulations to quantify dissolution dynamics. The results identify the Rayleigh number (Ra) as the key parameter controlling the transition from diffusion-dominated to convection-dominated dissolution. The onset time of convection and the maximum mass transfer rate are described by $t_{\text{onset}} = 1.45 \times 10^7 Ra^{-1.98}$ and $Sh_{\text{max}} = 0.045 Ra^{1.12}$, respectively. Simulations for South China Sea aquifers indicate that reservoir properties have limited influence on final dissolved CO₂ mass, but affect dissolution kinetics. These findings provide a calibrated basis for reservoir screening, injection design, and long-term assessment of offshore CO₂ sequestration.

Keywords CO₂ storage, Offshore saline aquifers, CO₂ diffusion coefficient, CO₂-enriched brine density, CO₂ convective dissolution, Scaling analysis

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1 Introduction

The continued rise in atmospheric CO₂ concentration remains a major driver of climate change, making deep decarbonization an urgent global priority [1, 2]. Although rapid deployment of renewable energy and electrification is essential, these measures alone are unlikely to eliminate emissions from hard-to-abate sectors in the near term [3–5]. Carbon capture and sequestration (CCS) is therefore widely regarded as a necessary component of net-zero pathways, particularly for large stationary sources such as power generation, cement, steel, and petrochemical industries [6, 7]. Among the available geological storage options, deep saline aquifers are considered especially attractive because of their large theoretical capacity, broad geographic distribution, and long-term storage potential [8–10].

In recent years, offshore CO₂ sequestration has received increasing attention because it can provide substantial additional capacity while reducing some of the land-use, groundwater protection, and public-acceptance constraints often associated with onshore projects [11, 12]. For coastal economies with concentrated industrial emissions, offshore storage also offers advantages in source–sink matching and infrastructure integration [13, 14]. In this context, the South China Sea is a strategically important target region due to its extensive sedimentary basins, reservoir–seal assemblages, and proximity to major emission clusters along the southern coast of China [15–17]. To support large-scale deployment in such offshore systems, it is essential to improve the reliability of predictions for CO₂ migration, dissolution, and long-term trapping [18–22].

A key mechanism controlling storage security in saline aquifers is solubility trapping (Fig. 1), whereby injected CO₂ dissolves into the resident brine [23, 24]. Compared

with buoyant free-phase CO₂, dissolved CO₂ is much less prone to remigration and leakage [25, 26]. More importantly, CO₂ dissolution commonly increases brine density, creating a gravitationally unstable boundary layer beneath the gas–brine interface [27, 28]. Once this dense layer becomes unstable, it can evolve from purely diffusive transport into density-driven convective mixing, greatly accelerating mass transfer from the free phase to the aqueous phase [29, 30]. This process is central to plume stabilization because it enhances dissolution rates, renews the interfacial concentration gradient, and shortens the timescale over which mobile CO₂ is converted into a more secure dissolved form [25]. Understanding the onset and evolution of density-driven convection is critical for evaluating both sequestration efficiency and long-term storage security [31–33].

Despite substantial progress in multiphase-flow and reactive-transport modeling [34–37], convective dissolution predictions remain sensitive to the fluid-property data used to parameterize the system [33, 38]. In many previous studies, key parameters such as the CO₂ diffusion coefficient in brine, CO₂ solubility, and the density of CO₂-enriched brine were estimated from empirical correlations or literature datasets obtained under simplified conditions [39–41]. Such treatments are often insufficient for offshore saline aquifers, where temperature, pressure, and salinity jointly affect molecular transport, phase equilibrium, fluid density, and ultimately the development of convective instability [42, 43]. In particular, the salinity dependence of CO₂ transport and the density increase induced by dissolution can significantly modify the Rayleigh number and therefore the onset time and intensity of convection. Neglecting these site-specific effects may lead to biased estimates of plume stabilization timescales and trapping performance [44–47].

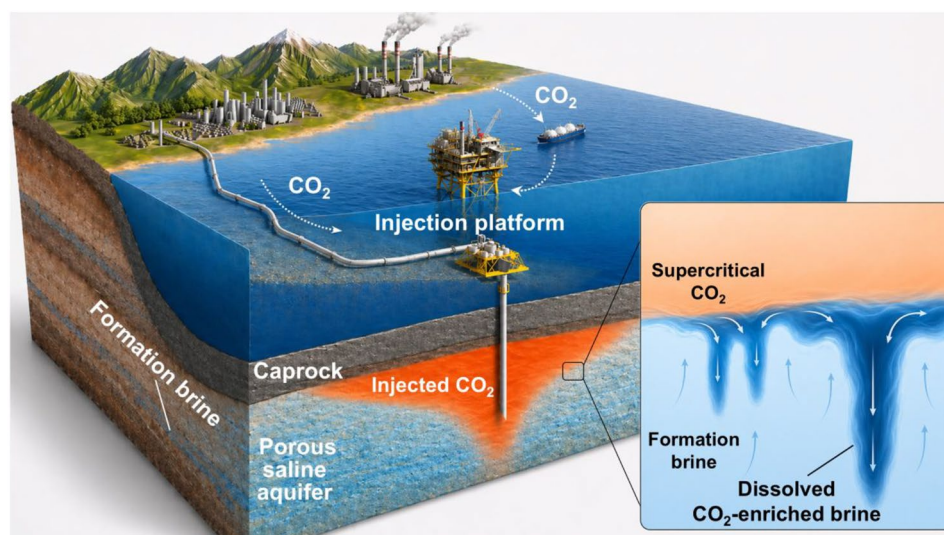


Fig. 1 Mechanisms of solubility trapping in offshore CO₂ sequestration

The uncertainty is compounded by limitations in experimental datasets under reservoir-relevant conditions [48, 49]. Traditional pressure-decay methods have been widely used to infer CO₂ diffusivity, but the measured values may be affected by boundary effects and density-driven natural convection, making it difficult to isolate the true molecular diffusion coefficient [43, 47, 50]. By contrast, Raman spectroscopy combined with capillary confinement can directly resolve concentration gradients while suppressing convective interference, offering a more robust route for determining molecular diffusivity under high-pressure, high-salinity conditions [51–53]. Similarly, accurate density measurements for CO₂-enriched brines remain relatively scarce, especially for actual reservoir-derived brines, even though density contrast is the immediate physical driver of convective instability [54, 55]. Therefore, a gap persists between fluids property characterization and CO₂ dissolution behavior prediction.

Against this background, the present study develops an integrated experimental–numerical framework to investigate the mechanisms and scaling laws of CO₂ convective dissolution in offshore saline aquifers relevant to the South China Sea. Unlike studies based primarily on synthetic brines or generalized correlations, this work uses actual reservoir-derived brine and determines the diffusion coefficient, solubility, and density of the CO₂–brine system from measurements under reservoir-representative conditions. Specifically, CO₂ diffusion coefficients are quantified using Raman spectroscopy–capillary method [56, 57], while the density of CO₂-enriched brine is determined using high-precision oscillating U-tube densitometry [58]. These experimentally constrained fluid properties are then incorporated into Darcy-scale numerical simulations to examine how molecular transport and dissolution-induced density contrast jointly govern plume evolution, convective onset, and mass transfer enhancement.

The focus of this study is the competition between buoyancy destabilization and diffusive stabilization, which can be expressed in terms of the Rayleigh number. By varying diffusion coefficient and density contrast in the numerical model, we seek to establish quantitative scaling laws for the onset time of convection and the maximum density-driven mass transfer intensity. In addition, representative engineering scenarios for the South China Sea are evaluated to clarify how reservoir temperature, pressure, and associated fluid properties affect dissolution kinetics and sequestration performance. Rather than considering storage capacity alone, the analysis emphasizes the practical importance of how rapidly injected CO₂ can be transferred into the dissolved phase, since accelerated dissolution directly improves storage security over Darcy-scale engineering timescales.

Overall, this study aims to bridge the gap between site-specific fluid property characterization and offshore CO₂ sequestration behavior prediction. The results provide a more physically grounded basis for reservoir screening, injection design, and long-term security assessment in the South China Sea, and is also relevant to other offshore saline aquifer systems, where dissolution trapping and convective mixing play important roles in CO₂ sequestration performance.

2 Materials and experimental characterization

This study combined in situ fluid property measurements with Darcy scale numerical simulations to investigate the mechanisms and scaling behavior of CO₂ convective dissolution in offshore saline aquifers. Reservoir derived brines from the South China Sea were used to measure the molecular diffusion coefficient of CO₂ and the density of CO₂-enriched brine under reservoir representative conditions. These experimentally constrained properties were then incorporated into a numerical model to quantify convective onset, mass transfer enhancement, and the effects of reservoir conditions on dissolution kinetics and storage performance.

2.1 Reservoir brine samples and experimental conditions

To obtain site-specific fluid-property data for offshore CO₂ storage in the South China Sea, a series of laboratory measurements was conducted using reservoir-derived brines under reservoir-representative pressure–temperature conditions. All experiments were performed at temperatures of 313.15–363.15 K and pressures of 7.5–17 MPa, corresponding to the typical in-situ conditions of deep saline aquifers in the South China Sea.

Formation-water samples were collected from two reservoir intervals, denoted as S1 and S2. Prior to measurement, the samples were subjected to the same collection, filtration, and compositional analysis procedures described in the source experimental study. The brines were dominated by Na⁺ and Cl[−], with measured salinities of 40.77 g/L for S1 and 35.63 g/L for S2. These actual reservoir-derived brines, rather than synthetic brines, were used throughout this study in order to preserve the site-specific fluid characteristics relevant to the South China Sea formation setting.

2.2 Determination of CO₂ diffusion coefficients

The molecular diffusion coefficient of CO₂ in brine was determined using Raman spectroscopy–capillary tube method, which enables direct observation of concentration evolution while suppressing density-driven convection through geometric confinement. In this method, CO₂-enriched brine was loaded into a fused-silica capillary sealed at one end and connected at the other end to a high-pressure syringe pump (Fig. 2a). The capillary tube

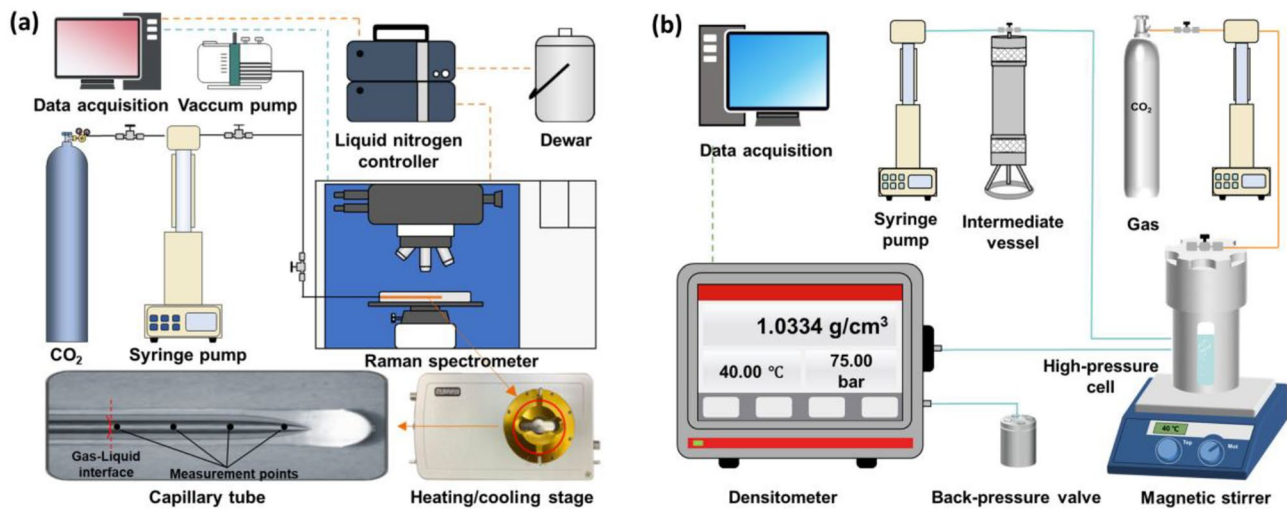


Fig. 2 Schematic of the experimental setup for measuring (a) CO₂ diffusion coefficient; (b) CO₂-enriched brine density

was placed on a temperature-controlled stage to maintain the target experimental conditions. Raman spectra were then collected sequentially at specified distances from the gas–liquid interface to obtain the transient spatial distribution of dissolved CO₂.

The fused-silica capillary used in the Raman diffusion experiment had an inner diameter of 300 μm and an effective length of approximately 10 cm. The small inner diameter makes the radial diffusion length scale much smaller than the axial diffusion length scale, allowing radial concentration gradients to be rapidly smoothed. The confined geometry also suppresses density-driven natural convection caused by CO₂ dissolution, supporting the approximation of one-dimensional axial diffusion in the capillary.

Because the radius of capillary tube was sufficiently small, radial concentration gradients were neglected and diffusion was treated as a one-dimensional process. The transport of dissolved CO₂ was therefore described by Fick's second law:

$$\frac{\partial C}{\partial t} = D \frac{\partial^2 C}{\partial z^2} \quad (1)$$

where C is the dissolved CO₂ concentration, t is time, z is the axial distance along the diffusion path, and D is the molecular diffusion coefficient.

Local CO₂ concentration was determined from the Raman intensity ratio between the CO₂ Fermi doublet and the O–H vibrational band of water, with water serving as an internal calibrator. Calibration curves relating Raman intensity ratio to CO₂ concentration were established using solubility values calculated from the Duan's model [44, 59] under the corresponding pressure–temperature conditions. Time-resolved concentration profiles were then fitted by finite-difference solutions of

the diffusion equation, and the diffusion coefficient was obtained by minimizing the deviation between the calculated and measured concentration fields. This approach allows the measured D to reflect true molecular diffusion, rather than an apparent effective diffusivity that includes convective contributions.

2.3 Measurement of CO₂-enriched brine density

The density of CO₂-enriched brine was measured using an oscillating U-tube densitometer (DMA 4200 M, Anton Paar, Austria) under the same pressure–temperature range as the diffusion experiments (Fig. 2b). Before each series of measurements, the instrument was calibrated with deionized water and nitrogen over the relevant thermodynamic range. Reference densities for the calibration fluids were obtained from NIST-traceable standards [60], and polynomial correction coefficients were generated to account for pressure and temperature effects on the oscillation signal. The experimental system and method used to measure the density of CO₂-enriched brine have disclosed in Chinese Patent CN118392713B [58], which also reports the relevant procedural details.

For each test, CO₂-saturated brine was transferred into an intermediate vessel and subsequently introduced into the densitometer cell under pressurized conditions. The system was allowed to equilibrate until both thermal and pressure stability were achieved, after which density was recorded. Repeated measurements were performed across the full range of temperature, pressure, and brine composition. The resulting dataset provides direct density values for CO₂-enriched brines and, by comparison with the corresponding CO₂-free brines, yields the density contrast ?? that drives dissolution-induced gravitational instability in the porous medium. Because density contrast is a key input to the Rayleigh number, these

measurements form the basis for physically constrained stability analysis in the numerical model.

3 Numerical simulation

3.1 Governing equations and model configuration

A two-dimensional Darcy-scale model was established using COMSOL Multiphysics to simulate CO₂ dissolution and density-driven convection in a saline aquifer. The objective was to quantify how molecular diffusion and dissolution-induced density increase jointly control the onset and evolution of convective mixing. This Darcy-scale configuration was chosen to isolate the effects of experimentally constrained fluid properties under controlled boundary conditions, rather than to reproduce the full plume evolution at a specific storage site. Site-scale simulations require detailed three-dimensional geological heterogeneity, well configuration, injection history, and multiphase-flow parameters, and are therefore reserved for future work using the scaling relations developed here. Fluid flow in the porous medium was governed by Darcy's law:

$$\mathbf{u} = -\frac{k}{\mu} \nabla p \quad (2)$$

where $\mathbf{u}$ is Darcy velocity, k is permeability, μ is dynamic viscosity, and p is pressure.

Transport of dissolved CO₂ was described by the convection–diffusion equation:

$$\frac{\partial C}{\partial t} + \mathbf{u} \cdot \nabla C = D \nabla^2 C \quad (3)$$

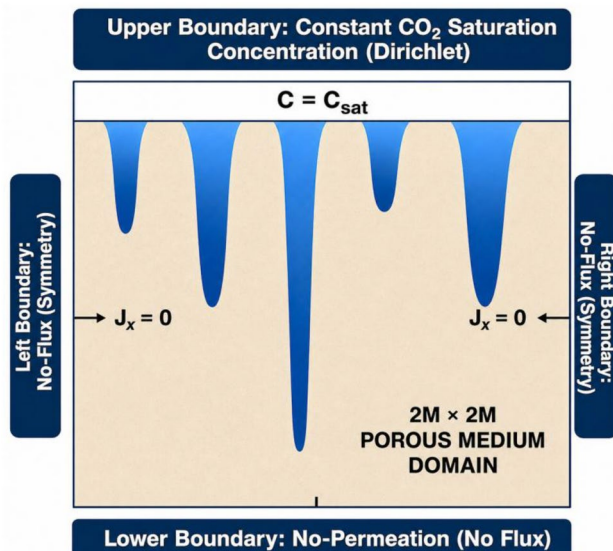


Fig. 3 Model size and boundary condition setup diagram

where C is dissolved CO₂ concentration and D is the molecular diffusion coefficient obtained from the experiments. The model domain was a 2 m × 2 m porous medium with a porosity of 0.2 and a permeability of 1×10^{-9} m², as shown in Fig. 3, consistent with the numerical simulation adopted for scaling analysis. To accurately resolve the concentration gradients and finger formation, the domain was discretized into 400 × 400 interior elements. Furthermore, specialized boundary layer meshes were applied along all four boundaries (top, bottom, and lateral), with each layer containing 2,000 elements to ensure high resolution at the interfaces. A grid independence test was performed to ensure that the chosen mesh density was sufficient to achieve converged solutions for both the onset time of convection and the mass transfer rate, with detailed verification results provided in the Supporting Information.

The upper boundary was prescribed as a constant dissolved-CO₂ concentration boundary to represent continuous contact with a CO₂-rich phase. The bottom boundary was treated as impermeable, and the lateral boundaries were specified as no-flow and no-flux boundaries. Under these conditions, a diffusive boundary layer develops beneath the top boundary and may become gravitationally unstable as the density of the CO₂-enriched brine increases.

The present two-dimensional Darcy-scale model is intended to resolve the local onset and development of density-driven convective fingers beneath a CO₂-brine interface under controlled boundary conditions, rather than to reproduce the full three-dimensional evolution of a field-scale CO₂ plume. This simplified configuration allows the effects of molecular diffusion coefficient D and density contrast $\Delta\rho$ on convective onset and mass-transfer enhancement to be isolated while maintaining sufficient numerical resolution of the diffusive boundary layer and finger structures. In real offshore saline aquifers, three-dimensional effects may influence finger spacing, finger coalescence, plume geometry, and the absolute mass-transfer rate. Therefore, the scaling relationships derived here should be interpreted as baseline local scaling relationships under the specified interface and boundary conditions. Field-scale applications require coupling these relationships with three-dimensional reservoir architecture, permeability heterogeneity, well configuration, and injection history.

3.2 Parameters and characteristic variables

The simulations were constrained by the experimentally measured fluid properties. In particular, the molecular diffusion coefficient and the density contrast between CO₂-enriched and unsaturated brine were taken directly from the in-situ measurements on actual reservoir-derived brines. In convective CO₂ dissolution, density

contrast provides the buoyancy driving force, diffusion controls boundary-layer development, and viscosity regulates fluid mobility and thereby the onset and intensity of convection. Accordingly, fluid viscosity was parameterized in the numerical model as a function of temperature, salinity, and dissolved CO₂ concentration. Brine viscosity was calculated using an empirical temperature-dependent correlation, and the dissolved CO₂ concentration was converted into mole fraction to correct the viscosity of CO₂-free brine, yielding the viscosity of CO₂-enriched brine [61]. Detailed formulations of the viscosity model are provided in the Supporting Information. Together, these treatments ensure that the simulations are constrained by measured reservoir-fluid properties rather than relying solely on generalized empirical correlations.

The tendency for density-driven instability was characterized by the Rayleigh number, Ra :

$$Ra = \frac{kg\Delta\rho H}{\mu D} \quad (4)$$

where g is gravitational acceleration, H is the characteristic height of the domain. In this study, Ra represents the competition between buoyancy destabilization and diffusive stabilization and serves as the principal dimensionless parameter for subsequent scaling analysis.

In Eq. (4), μ denotes the viscosity of CO₂-bearing brine under the corresponding pressure-temperature-salinity condition, rather than a constant pure-brine viscosity. The viscosity variation associated with reservoir conditions and dissolved CO₂ were therefore considered in the Rayleigh-number calculation. The viscosity values used for the representative South China Sea storage scenarios are listed in Table 2.

To quantify the transition from diffusion- to convection-dominated transport, the onset time of convection (t_{onset}) was determined using a velocity-based threshold. The transition to the convective regime is identified by the inflection point in the maximum migration velocity (U_{max}) of the CO₂-enriched brine interface, marking the shift from diffusion-controlled attenuation to convection driven growth [62–64]. To quantify the solute accumulation during the pre-instability phase, a diffusive length scale was evaluated based on classical diffusion theory. Prior to the onset of convection, the critical equivalent height of the dissolved CO₂ boundary layer is assumed to

Table 1 Simulation conditions for the mechanistic analysis of density driven convection

Parameter	Unit	Numerical value	Source
Density contrast, $\Delta\rho$	kg/m ³	0.1, 12.1, 17.3, 26.6	Experimental parameters
Density contrast, $\Delta\rho$	kg/m ³	8.0, 14.5, 22.0, 31.0, 36.0	Extended parameters
Diffusion coefficient, D	10 ⁻⁹ m ² /s	2.89, 3.53, 4.21, 4.78	Experimental parameters
Diffusion coefficient, D	10 ⁻⁹ m ² /s	1.57, 2.14, 5.49, 6.12	Extended parameters

scale with the characteristic diffusion length. This scaling relationship serves as a theoretical benchmark for comparing the simulated solute-layer thickness with the diffusion length at the onset of instability [65–67].

The Sherwood number Sh representing the ratio of convective mass transfer to pure molecular diffusion, was employed to quantify the enhancement of the dissolution process. Specifically, the peak Sherwood number (Sh_{peak}) was identified to establish scaling laws with the Rayleigh number.

$$Sh(t) = \frac{\int_{\Gamma} \mathbf{n} \cdot \mathbf{N}_c d\Gamma / H}{W \phi_m D c_{eq}} \quad (5)$$

3.3 Simulation cases and engineering scenarios

Two groups of simulation cases were considered in this study. The first group (Table 1) was designed for mechanistic analysis. The diffusion coefficient and density contrast were varied to isolate their individual and combined effects on plume evolution, convection onset, and mass transfer intensity under fixed conditions of 318.15 K and 13.8 MPa, while the density of CO₂ free brine and brine viscosity were kept constant at 1003.28 kg/m³ and 8.9769 × 10⁻⁴ Pa s, respectively. These simulations provided the basis for establishing scaling relationships between the Rayleigh number and the characteristic parameters of density driven convection.

The second group was used for engineering-oriented evaluation of representative CO₂ sequestration conditions of South China Sea. In these cases, reservoir temperature and pressure were varied to generate corresponding changes in CO₂ solubility, density, diffusivity, and viscosity (Table 2). The resulting parameter combinations were then used to assess how reservoir conditions influence dissolution kinetics, the time required to

Table 2 Fluid properties and reservoir conditions of the CO₂ brine system for representative South China Sea storage scenarios

Case	T (°C)	P (MPa)	Solubility (mol/L)	CO ₂ molar fraction	ρ (kg/m ³)	D (10 ⁻⁹ m ² /s)	μ (10 ⁻⁴ Pa·s)
1	52.4	7.50	0.8126	0.01456	1026.6	2.890	6.7850
2	66.9	11.6	0.9024	0.01636	1017.3	3.528	5.4294
3	80.0	14.5	0.9113	0.01661	1012.1	4.213	4.5564
4	90.0	17.0	0.9397	0.01735	1000.1	4.775	4.0413

reach high fractions of ultimate dissolved storage, and the balance between rapid solubility trapping and long-term sequestration performance. This hierarchical design enabled the model to address both the fundamental transport mechanisms and their practical implications for offshore CO₂ sequestration.

4 Results and discussion

4.1 In-situ characterization of CO₂-brine properties

The reliability of the numerical analysis depends directly on the accuracy of the fluid-property data used for model parameterization. The key transport and stability parameters governing CO₂ convective dissolution, namely the molecular diffusion coefficient and the density of CO₂-enriched brine, were determined experimentally using actual reservoir-derived brines under South China Sea-representative conditions. These measurements provide a

physically constrained basis for evaluating density-driven convection in offshore saline aquifers.

Figure 4 presents the Raman-based calibration and the time-resolved evolution of dissolved CO₂ concentration in the capillary system. The Raman peak intensity ratio between the CO₂ Fermi doublet and the water band exhibits a stable linear relationship with dissolved CO₂ concentration over the investigated pressure–temperature range, confirming the robustness of the calibration procedure. During diffusion, the intensity of the CO₂-related peak increases progressively with time, whereas the water band remains essentially unchanged. This behavior is consistent with a diffusion-dominated transport process within the capillary and indicates that the experimental geometry effectively suppresses convective disturbance.

The molecular diffusion coefficients of CO₂ measured in South China Sea formation brines range from

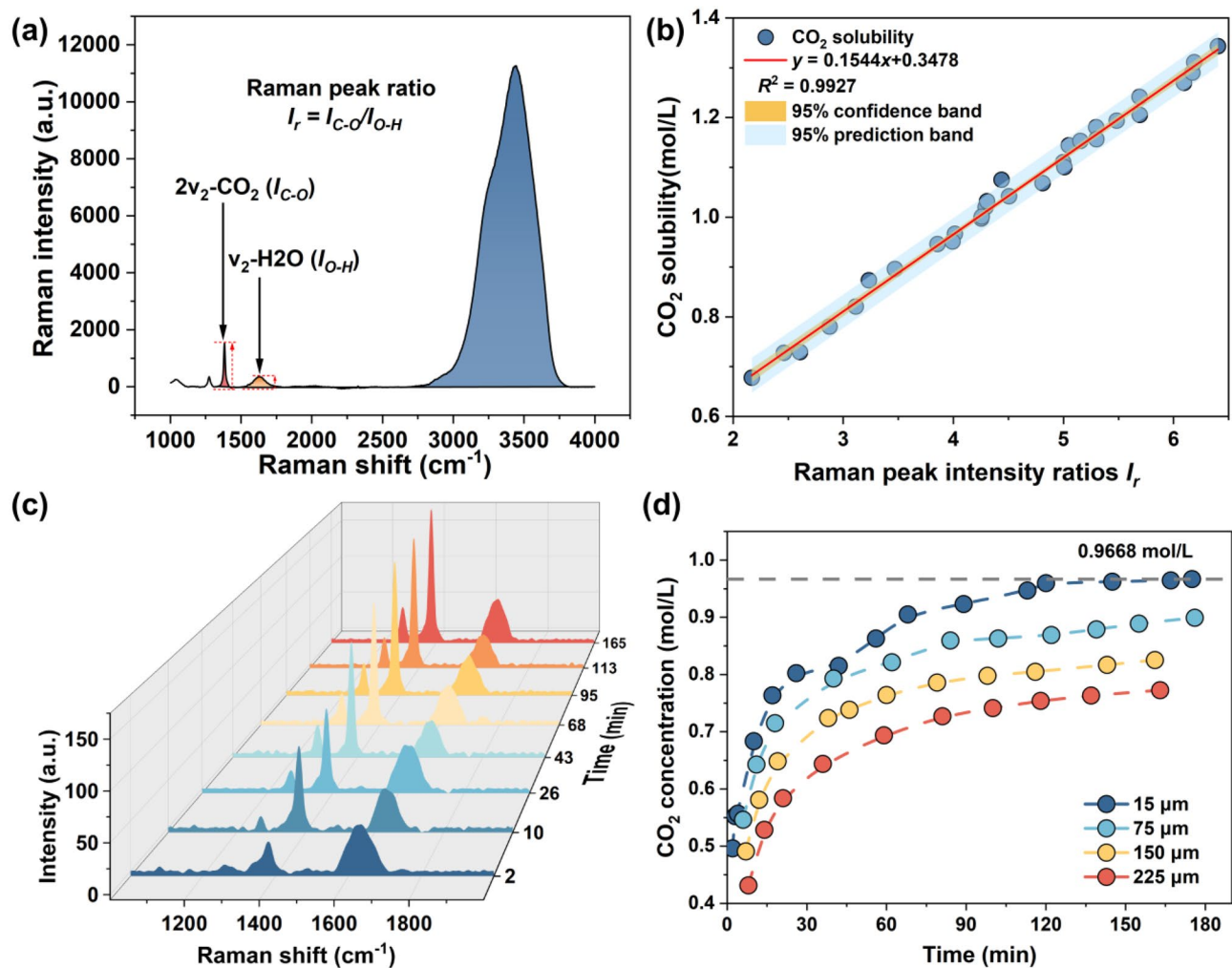


Fig. 4 Raman-based quantification of dissolved CO₂ in brine: (a) representative Raman spectrum of CO₂-enriched water; (b) calibration relationship between Raman peak intensity ratio and CO₂ concentration; (c) evolution of Raman spectra with time; (d) concentration evolution at different distances from the gas–liquid interface

2.89×10^{-9} to 4.78×10^{-9} m²/s at temperatures of 313.15 to 363.15 K and pressures of 7.5 to 17 MPa. As shown in Fig. 5, temperature exerts the strongest control on diffusivity, with D increasing nearly linearly as temperature rises. This trend can be attributed to enhanced molecular mobility and reduced solution viscosity at elevated temperatures. Within the investigated pressure range of 7.5–17 MPa, pressure had only a weak influence on D , with the total variation remaining below approximately 7%. However, the present data are insufficient to determine whether this weak pressure dependence persists at pressures exceeding 30 MPa. At higher pressures, changes in CO₂ dissolution behavior, fluid viscosity, and molecular interactions may alter the pressure dependence of diffusivity. Therefore, direct extrapolation of the present pressure–diffusivity relationship beyond the experimental range is not recommended, and additional measurements are required under higher-pressure reservoir conditions. It should be emphasized that this weak pressure dependence was observed only within the investigated pressure range of 7.5–17 MPa. The present data are insufficient to determine whether the same trend persists at pressures exceeding 30 MPa, where changes in CO₂ dissolution behavior, fluid viscosity, and molecular interactions may alter the pressure dependence of diffusivity. Therefore, direct extrapolation of the present pressure–diffusivity relationship beyond the experimental range is not recommended, and additional measurements are required under higher-pressure reservoir conditions. This limited pressure dependence differs from many values reported using pressure-decay approaches, which often reflect an apparent diffusivity that includes contributions from natural convection rather than pure molecular diffusion.

The distinction between molecular diffusivity and the apparent effective diffusivity inferred from

pressure-decay experiments is particularly important for Darcy-scale simulations. Because density-driven flow is typically resolved explicitly through the momentum equation, the diffusion term should be parameterized using molecular diffusivity, while buoyancy-driven mixing is captured by the resolved flow field. Otherwise, convection-enhanced transport may be represented in both the advection and diffusion terms, leading to overestimated dissolution rates. As shown in Fig. 6(a), the apparent effective diffusivities reported in previous pressure-decay studies are consistently higher than the molecular diffusivities measured here under comparable pressure conditions, typically by several-fold and in some cases approaching one order of magnitude. This discrepancy arises because the pressure-decay-derived values reflect not only molecular diffusion but also density-driven natural convection, whereas the Raman–capillary measurements reported here were obtained under convection-suppressed conditions.

Figure 6(b) further shows that salinity suppresses molecular diffusivity. S1 has a higher salinity (40.77 g/L) than S2 (35.63 g/L), which increases ionic strength and solution viscosity and therefore tends to reduce CO₂ molecular diffusivity. Since most diffusion measurements were conducted using S1, the reported diffusion coefficients mainly represent the higher-salinity reservoir-brine condition. Increasing ionic strength reduces CO₂ solubility through the salting-out effect and increases solution viscosity, jointly hindering diffusive transport. This trend underscores the importance of using site-specific formation brines in subsurface CO₂ storage simulations, as neglecting salinity effects may overestimate early-stage dissolution rates and underestimate the time required for density contrasts to develop, thereby

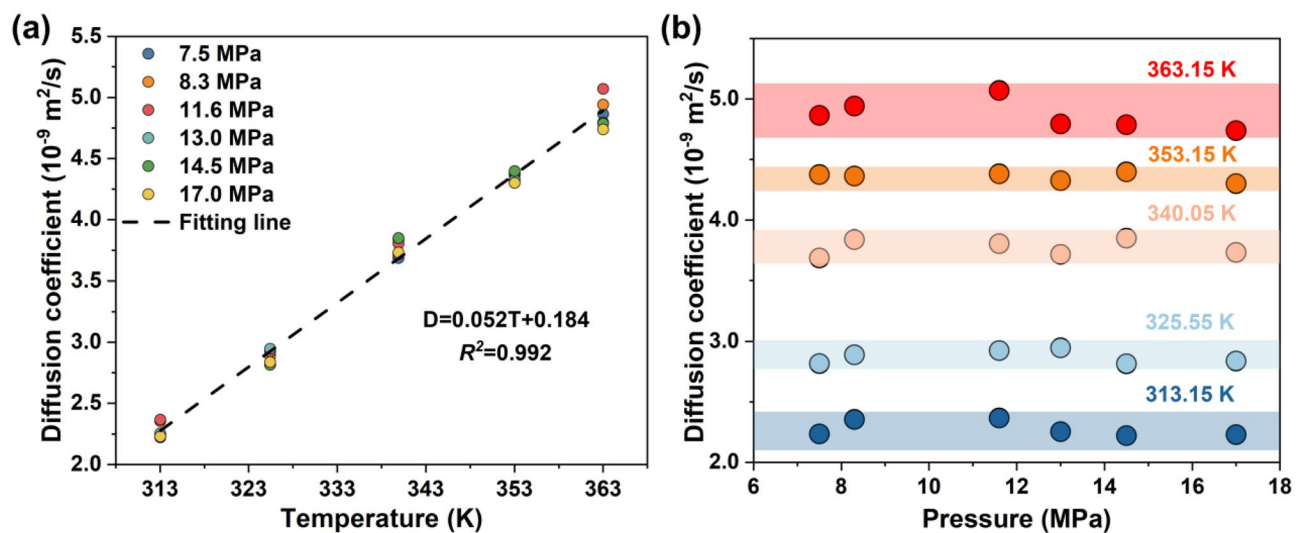


Fig. 5 Diffusion coefficient of CO₂ in South China Sea formation brine as a function of (a) temperature and (b) pressure

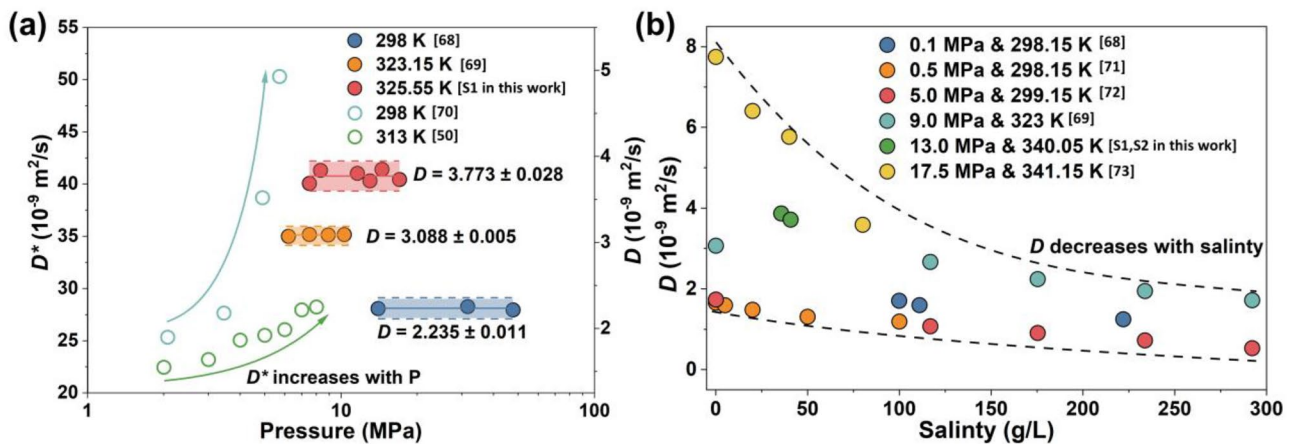


Fig. 6 Diffusivity characteristics of CO_2 in brine. **(a)** Comparison between pressure decay derived apparent effective diffusivity (D^*) and molecular diffusivity as a function of pressure. Data sources: Cadogan, Hallett [68]; Perera, Deng [69]; S1 in this work; Ahmadi, Jamialahmadi [70]; Li, Qiao [50]; **(b)** Dependence of the molecular diffusion coefficient on salinity. Data sources: Cadogan, Hallett [68]; Zhang, Geng [71]; Sell, Fadaei [72]; Perera, Deng [69]; S1 and S2 in this work; Zarghami, Boukadi [73]

affecting predictions of convective onset and subsequent mass-transfer behavior.

The measured densities of CO_2 -enriched brines further clarify the origin of buoyancy-driven instability. Figure 7(a, b) shows that the density of CO_2 -enriched brine increases approximately linearly with pressure and decreases with temperature. These trends reflect the combined effects of pressure-enhanced CO_2 dissolution, liquid compressibility, and thermal expansion. Higher pressure promotes dissolution and compresses the liquid phase, thereby increasing density, whereas higher temperature lowers CO_2 solubility and enhances thermal expansion, resulting in lower density. Comparison among deionized water, S1, and S2 in Fig. 7(c, d) shows that saline solutions are consistently denser than deionized water under comparable conditions, confirming that salinity elevates the baseline aqueous-phase density. Accordingly, the higher salinity of S1 results in a higher baseline brine density than that of S2 under comparable pressure-temperature conditions. Although salinity tends to reduce CO_2 solubility, the measured densities indicate that CO_2 dissolution still modifies fluid density sufficiently to generate a buoyancy contrast relevant to convective instability. This density contrast provides the gravitational driving force for destabilizing the diffusive boundary layer and initiating density-driven convection. Accordingly, the density measurements constrain a key thermodynamic input for assessing the onset and strength of convective mixing.

Taken together, the experimental results show that the South China Sea reservoir brines exhibit three key features relevant to offshore CO_2 storage: molecular diffusivity is strongly temperature-dependent but only weakly pressure-dependent; salinity suppresses CO_2 diffusion; and dissolution always increases brine density sufficiently

to induce gravitational instability. These experimentally determined fluid properties provide the foundation for the subsequent numerical analysis of convective dissolution.

4.2 Spatiotemporal evolution of density-driven convection

The experimentally constrained numerical simulations show that CO_2 dissolution in saline aquifers evolves through several transport regimes. At the initial stage, dissolved CO_2 accumulates beneath the upper boundary and forms a relatively uniform diffusive boundary layer. During this period, mass transfer is controlled mainly by molecular diffusion, and the concentration field remains laterally smooth. As dissolution continues, the CO_2 -enriched layer becomes denser and progressively destabilized by gravity. When buoyancy overcomes diffusive stabilization, the interface loses stability and downward fingers develop, indicating the transition to a convection dominated regime.

Figure 8 presents the evolution of the dissolved CO_2 concentration field under different molecular diffusion coefficients at a fixed density contrast. The selected cases correspond to $D = 2.89, 3.53, 4.21$, and $4.78 \cdot 10^{-9} \text{ m}^2/\text{s}$. At $t = 0.5$ days, all cases exhibit a laterally continuous diffusive boundary layer beneath the upper boundary, indicating that mass transfer is dominated by molecular diffusion. At $t = 5$ days, several distinct fingers are visible, and their vertical development remains limited. At $t = 10$ days, the effect of D becomes evident. For $D = 2.89 \cdot 10^{-9} \text{ m}^2/\text{s}$, the fingers with a penetration depth of about 50–60% of the domain height. As D increases to $4.78 \cdot 10^{-9} \text{ m}^2/\text{s}$, the fingers penetration depth decreases. This comparison shows that increasing D suppresses the amplification of interfacial perturbations during the early

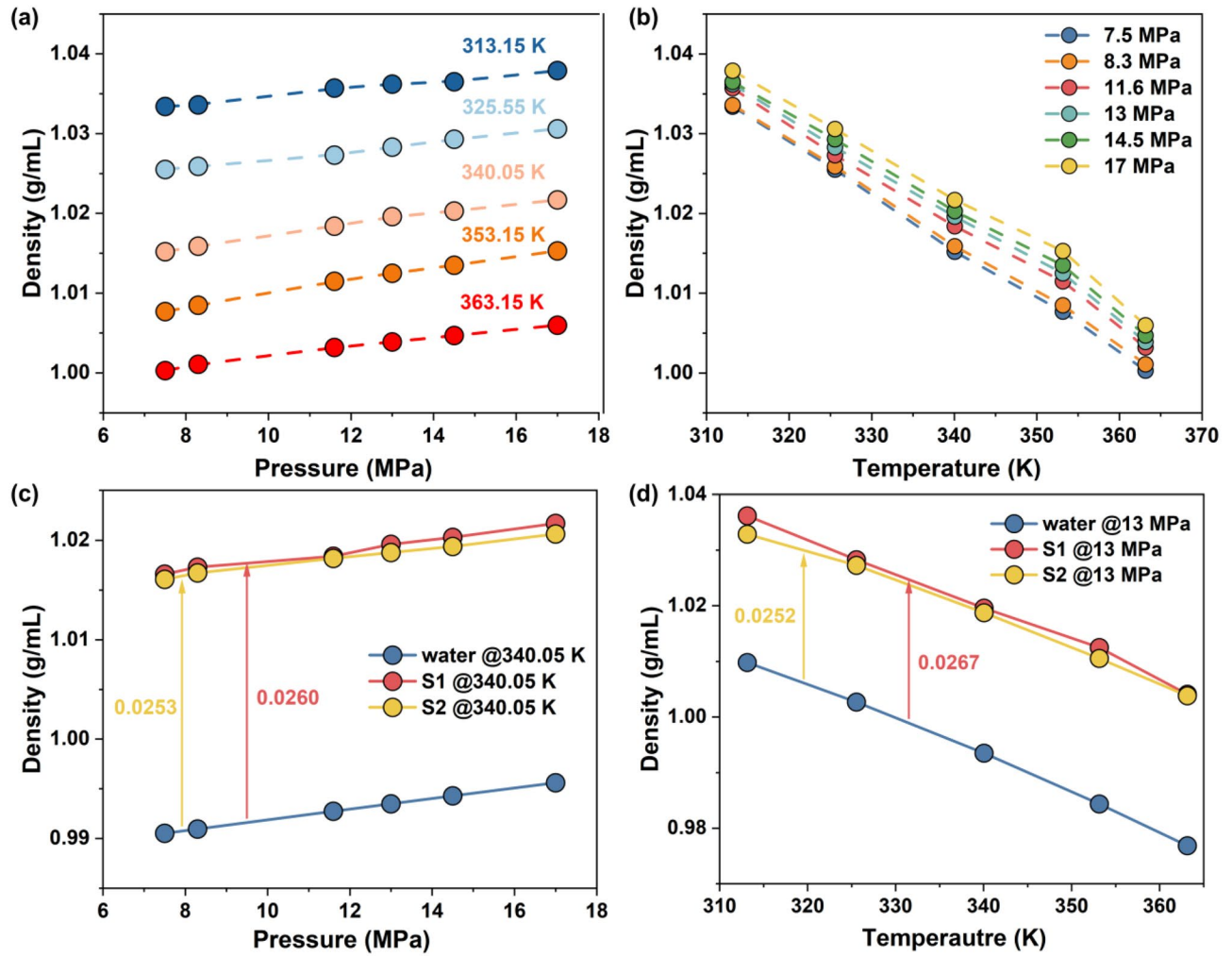


Fig. 7 Density of CO₂-enriched brines as a function of (a) pressure and (b) temperature; comparison of density among deionized water, S1, and S2 at (c) fixed temperature and (d) fixed pressure

development stage. At $t = 90$ days, all cases have evolved into a convection developed state, and the dissolved CO₂ has spread through most of the domain. The concentration field is then characterized less by isolated finger tips and more by the extent of vertical homogenization.

Following the analysis of molecular diffusion effects, Fig. 9 further shows the evolution of the dissolved CO₂ concentration field under different density contrasts at a fixed diffusivity of $D = 2.89 \times 10^{-9} \text{ m}^2/\text{s}$. The selected cases correspond to $\Delta\rho = 0.1, 12.1, 17.3$, and 26.6 kg/m^3 . Compared with Fig. 8, variation in $\Delta\rho$ produces a more direct change in instability intensity. At $t = 5$ days, the case with $\Delta\rho = 0.1 \text{ kg/m}^3$ remains almost entirely diffusion dominated, with a flat concentration interface and no visible fingering, whereas clear descending fingers have already formed when $\Delta\rho$ reaches 12.1 kg/m^3 . As $\Delta\rho$ further increases to 17.3 and 26.6 kg/m^3 , the fingers extend more deeply into the domain. At $t = 10$ days, the penetration depth is still negligible for $\Delta\rho = 0.1 \text{ kg/m}^3$, but increases

to about 60% of the domain height for $\Delta\rho = 12.1 \text{ kg/m}^3$, about 70% to 80% for $\Delta\rho = 17.3 \text{ kg/m}^3$, and exceeds 90% for $\Delta\rho = 26.6 \text{ kg/m}^3$.

The number of fingers also increases markedly, while the lateral spacing becomes smaller. By $t = 90$ days, the low $\Delta\rho$ case is still characterized mainly by diffusive thickening, whereas the higher $\Delta\rho$ cases have developed established convective structures with downward transport. These results indicate that increasing $\Delta\rho$ promotes the amplification of interfacial perturbations and accelerates the transition from diffusion-controlled transport to convection-controlled transport. In comparison with molecular diffusion coefficient, the effect of density contrast on finger initiation and vertical development is more pronounced, confirming that buoyancy generated by CO₂ dissolution is the primary control on density driven convection in the present system.

The flow fields corresponding to the concentration evolution are shown by the pressure and streamline

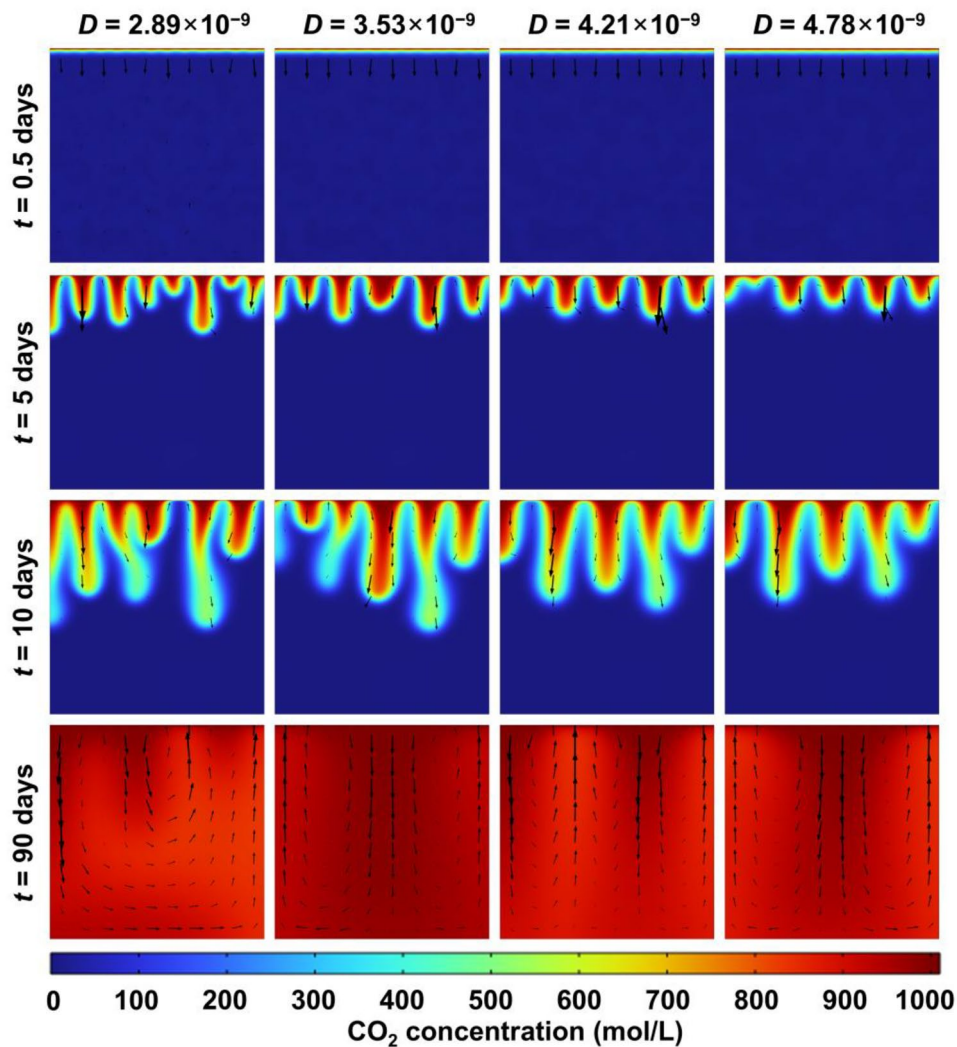


Fig. 8 Temporal evolution of dissolved CO_2 concentration fields under different molecular diffusion coefficients at $\Delta\rho = 12.1 \text{ kg/m}^3$. Higher D smooths the concentration boundary layer and delays early finger development

distributions in Figs. 10 and 11. At the early stage, when dissolution is mainly controlled by molecular diffusion, streamlines are not clearly developed. As the dense boundary layer becomes unstable, localized circulation cells gradually appear. CO_2 rich brine descends from the upper boundary in narrow flow paths, while relatively fresh brine rises in the surrounding region toward the dissolution interface. This return flow continuously renews the fluid near the upper boundary, sustains the concentration difference across the interface, and promotes continued dissolution. As a result, the coupling between dissolution, density increase, and fluid circulation enhances mass transfer once convection is established.

The interplay between the results in Figs. 10 and 11 further elucidates that the flow structure is governed by the competing effects of density-driven destabilization and diffusion-induced stabilization. Under a larger density

contrast, circulation is more developed, and the descending plumes penetrate deeper into the domain. Under a larger molecular diffusion coefficient, the streamline pattern develops more slowly. These results indicate that density contrast promotes the growth of convective instability, whereas molecular diffusion suppresses it by smoothing the concentration boundary layer. The combined effect of these two factors can be described by the Rayleigh number. The spatiotemporal evolution of concentration and flow fields therefore provides the physical basis for the scaling analysis in the next subsection.

4.3 Scaling laws for convective onset and mass-transfer enhancement

To establish predictive relationships for density driven convection, the temporal evolution of flow and mass transfer was analyzed as a function of the Rayleigh number. Figure 12 presents the time histories of maximum

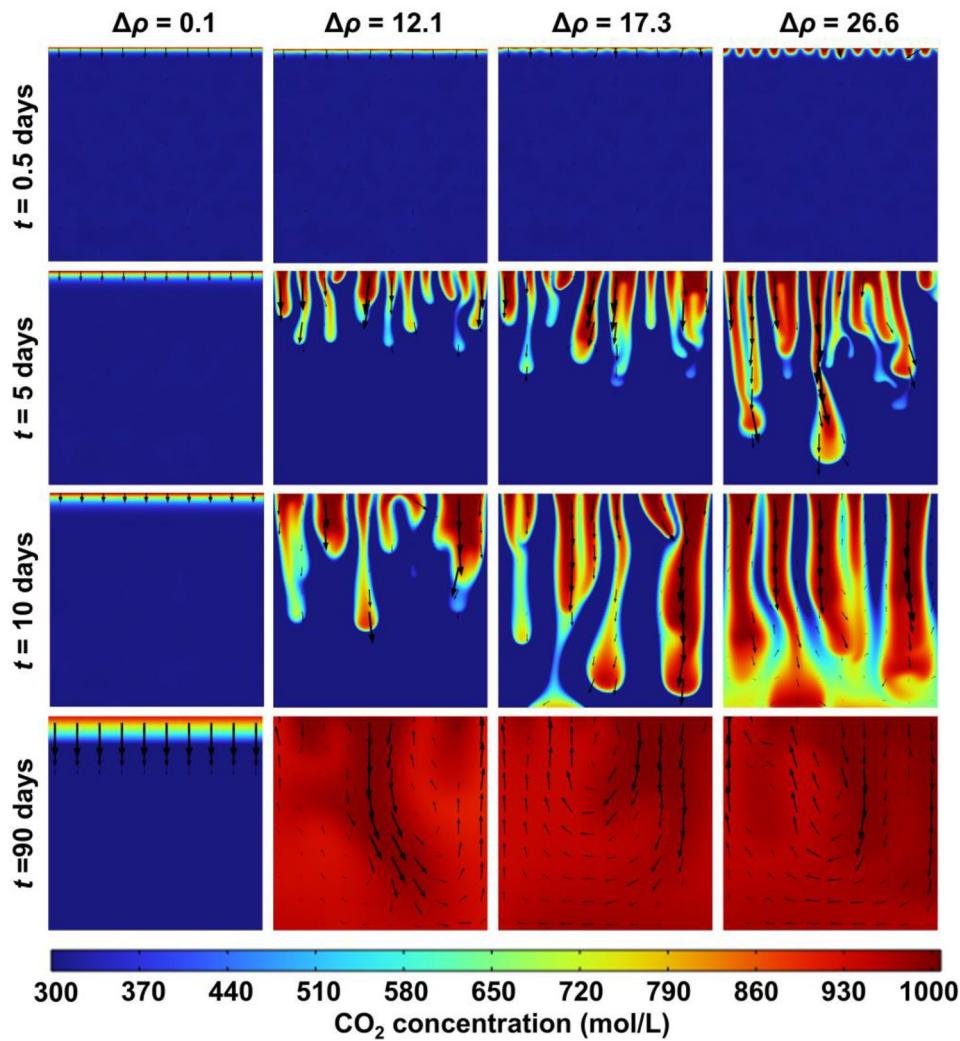


Fig. 9 Temporal evolution of dissolved CO_2 concentration fields under different density contrasts at $D=2.89 \times 10^{-9} \text{ m}^2/\text{s}$. Increasing $\Delta\rho$ strengthens buoyancy-driven fingering and accelerates downward CO_2 -rich brine migration

velocity and Sherwood number for representative cases with different diffusion coefficients and density contrasts. Although the absolute magnitudes differ, all cases exhibit a similar evolution pattern, including an initial diffusion-controlled stage, a convection development stage, and a late-stage decay as the concentration gradient progressively weakens.

As illustrated in Fig. 12(a), the velocity curve remains at a relatively low magnitude during the initial stage, followed by a rapid surge upon the onset of convection, and eventually decays gradually toward zero at later times. Under the same $\Delta\rho$, the maximum velocity varies only within 0.89×10^{-6} to $0.99 \times 10^{-6} \text{ m/s}$, indicating that changes in D mainly alter the timing and temporal development of convection, while their effect on the overall magnitude of flow intensity is comparatively limited. A similar trend is observed for the Sherwood number in Fig. 12(c). The peak Sherwood number Sh_{peak} ranges only

from about 100 to 200, with the highest value occurring in the case with the relatively lower D . This trend is consistent with the concentration evolution in Fig. 8, where lower D leads to earlier fingering, stronger downward penetration, and more effective interfacial renewal. As D increases, the convective growth stage is delayed, and the corresponding mass transfer peak becomes weaker.

The effect of density contrast is more pronounced compared with diffusion coefficient. As shown in Fig. 12(b), increasing $\Delta\rho$ causes a clear increase in maximum velocity over the early and intermediate stages. The maximum velocity rises to about $2.1 \times 10^{-6} \text{ m/s}$ at $\Delta\rho = 26.6 \text{ kg/m}^3$, which is more than twice the maximum velocity range observed for the cases with different diffusion coefficients. Figure 12(d) shows the same trend for the Sherwood number. The peak value increases from about 90 at $\Delta\rho = 0.1 \text{ kg/m}^3$ to about 1300 at $\Delta\rho = 26.6 \text{ kg/m}^3$. At the same time, the decline occurs earlier for larger $\Delta\rho$,

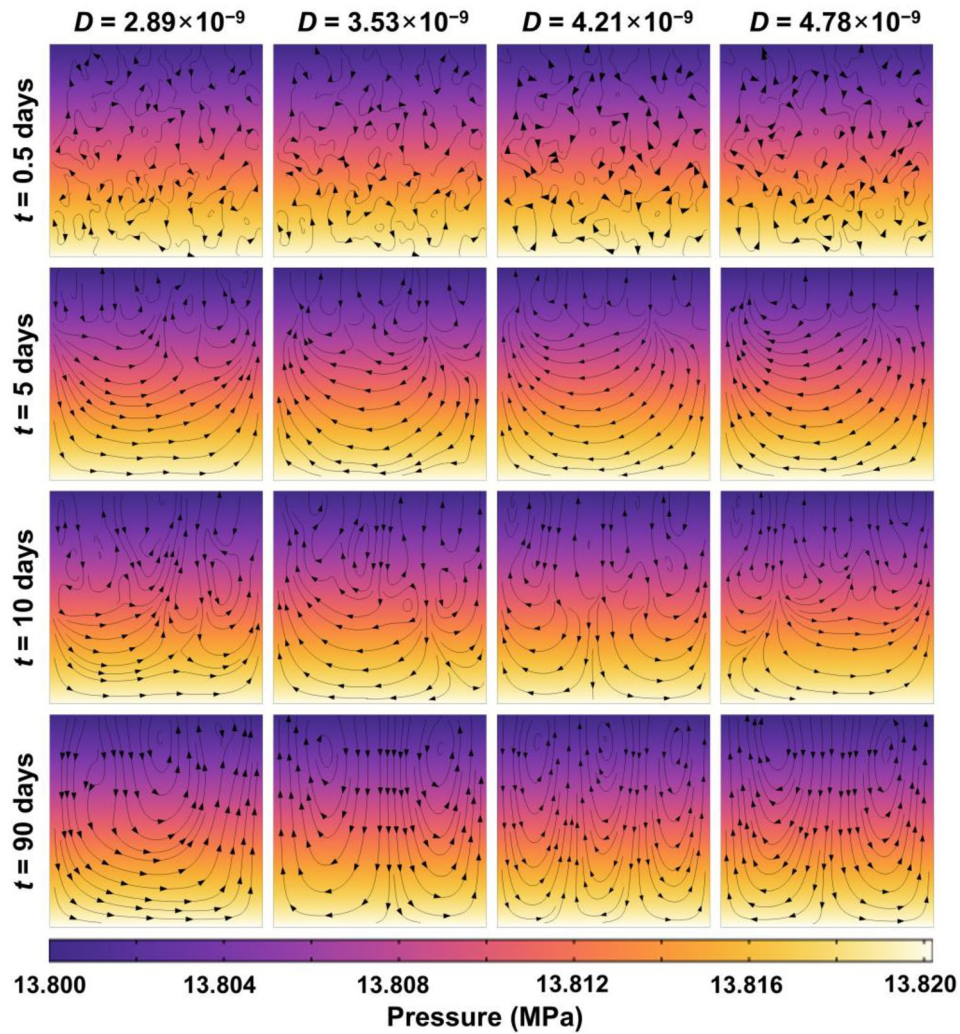


Fig. 10 Pressure and streamline fields under different molecular diffusion coefficients at $\Delta\rho = 12.1 \text{ kg/m}^3$. Higher D delays the formation of circulation cells by weakening early concentration gradients

indicating faster development and earlier exhaustion of the unstable boundary layer. Compared with the diffusion coefficient cases, variation in density contrast produces a much stronger response in both flow intensity and dissolution rate. This result agrees with the concentration and streamline fields in Figs. 9 and 11, which show earlier finger initiation and stronger plume penetration at larger $\Delta\rho$.

The onset of convection is identified as the initial slope deviation (inflection point) in the temporal evolution of $U_{\max}$, as depicted in Fig. 13(a). The corresponding values of maximum velocity, onset time, and Rayleigh number are summarized in Table 3 for the cases with different density contrasts. Figure 13(b) shows that t_{onset} decreases with increasing Ra . Over the investigated range, Ra increases from about 3024 to 13,608, whereas t_{onset} decreases from about 0.145 to 0.020 days. Regression of the simulation results gives,

$$Sh(t) = \frac{\int_{\Gamma} \mathbf{n} \cdot \mathbf{N}_c d\Gamma / H}{W \phi_m D c_{eq}} \quad (6)$$

This relation indicates an approximately inverse square dependence of t_{onset} on Ra . It shows that the transition from diffusion-controlled transport to convection occurs much earlier as buoyancy increasingly exceeds diffusive and viscous resistance. The fitted exponent is close to 2, and the data remain closely distributed around the fitted line, confirming that Ra is the main parameter controlling the onset of instability in the present system.

The fitted exponent of -1.98 should be interpreted as an empirical scaling exponent under the present finite-domain Darcy-scale configuration rather than as a universal linear-stability exponent. Unlike classical linear stability analyses based on idealized assumptions such as infinitesimal perturbations, simplified boundaries, and constant fluid properties, the present relation was

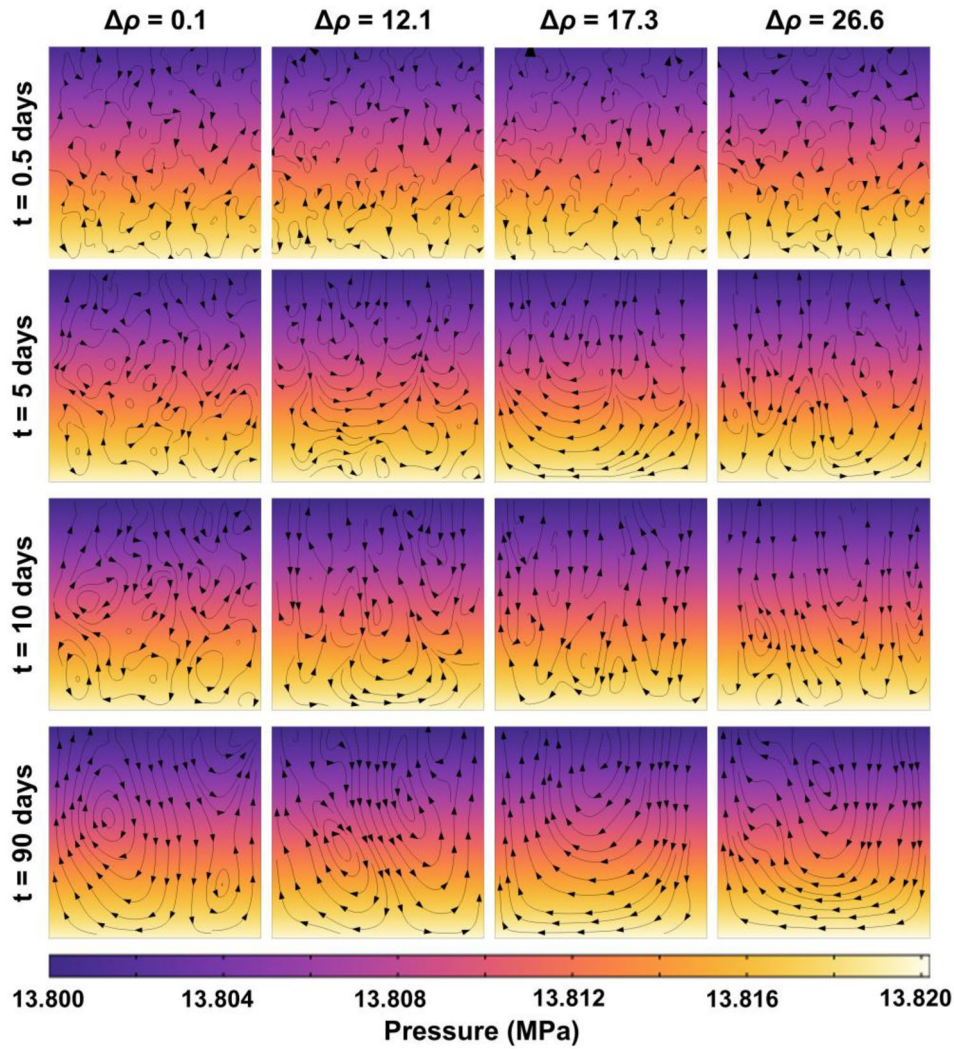


Fig. 11 Pressure and streamline fields under different density contrasts at $D = 2.89 \times 10^{-9} \text{ m}^2/\text{s}$. Larger $\Delta\rho$ produces stronger circulation cells and deeper downward transport

obtained from transient simulations with experimentally constrained D and $\Delta\rho$ and a velocity-based onset criterion. Physically, higher Ra strengthens buoyancy relative to viscous and diffusive damping, thereby accelerating the onset of gravitational instability within the CO_2 -enriched boundary layer. The near inverse-square dependence therefore indicates the strong sensitivity of t_{onset} to the buoyancy-diffusion competition within the investigated parameter range.

The enhancement of dissolution after convection onset was evaluated using the peak Sherwood number. The Sh_{peak} and Ra under various density differences and diffusion coefficients are summarized in Table 4.

As shown in Fig. 13(b), the Sh_{peak} increases with Ra according to:

$$Sh_{\text{peak}} = 0.045Ra^{1.12} \quad (7)$$

with an R^2 of 0.965. In the investigated range, the Sh_{peak} increases from about 257 at $Ra = 2295$ to about 1317 at $Ra = 10,055$. The exponent slightly above unity indicates that mass transfer enhancement becomes progressively stronger with increasing Ra . This trend reflects stronger circulation, deeper finger penetration, and more effective renewal of the fluid near the dissolution interface at larger Ra . As a result, the convective contribution to dissolution becomes increasingly important as the system departs from the diffusion-controlled regime.

Further insight into the convection onset process is provided by the evolution of the diffusive boundary layer prior to instability. Figure 14(a) gives the dissolved CO_2 mass at convective onset time, which is used to determine the equivalent mass thickness. The relevant parameters for various diffusion coefficient scenarios are summarized in Table 5. As D increases, both the dissolved CO_2 mass and the equivalent height H_{eq} at t_{onset}

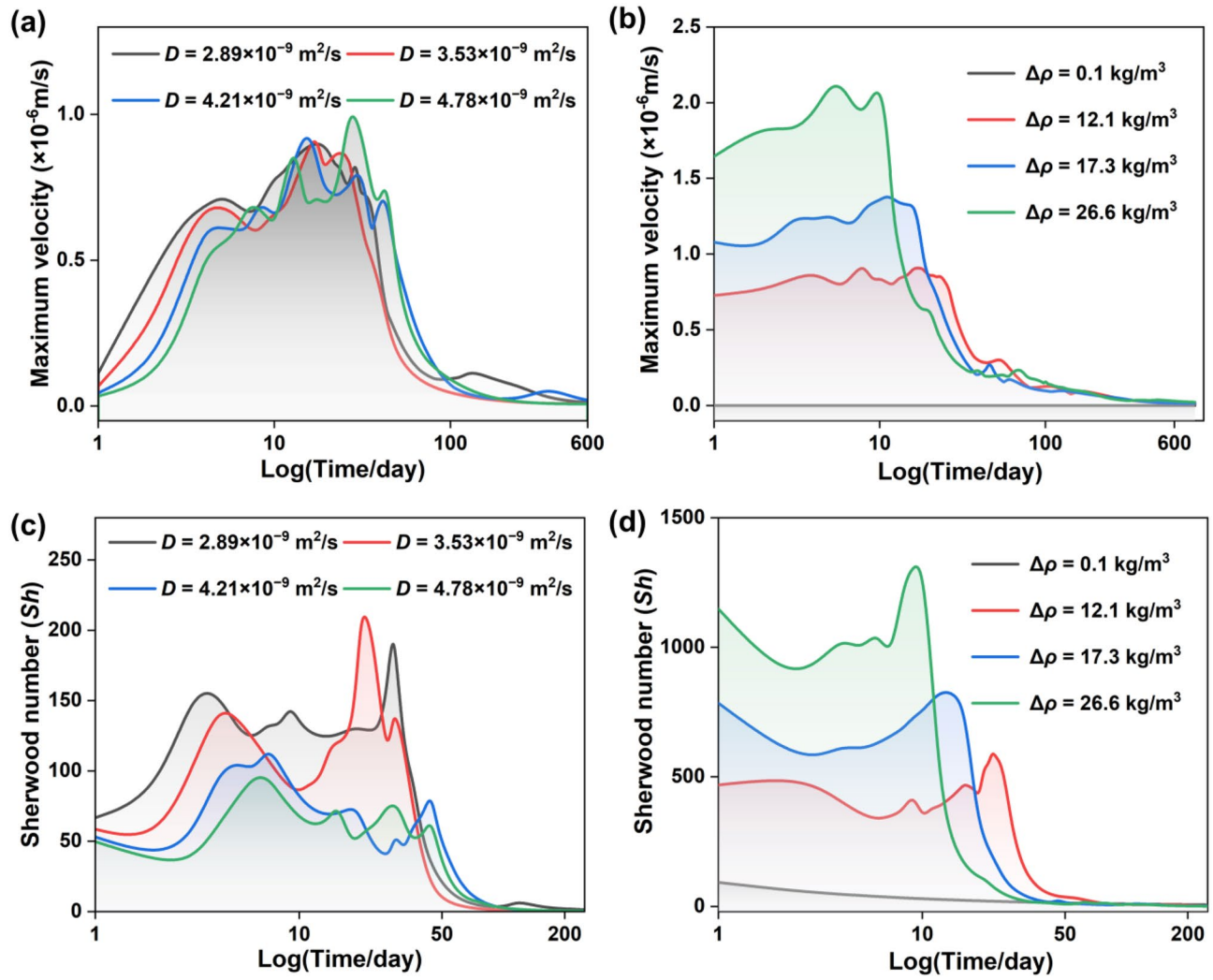


Fig. 12 Temporal evolution of maximum velocity and Sherwood number for representative cases with different diffusion coefficients and density contrasts

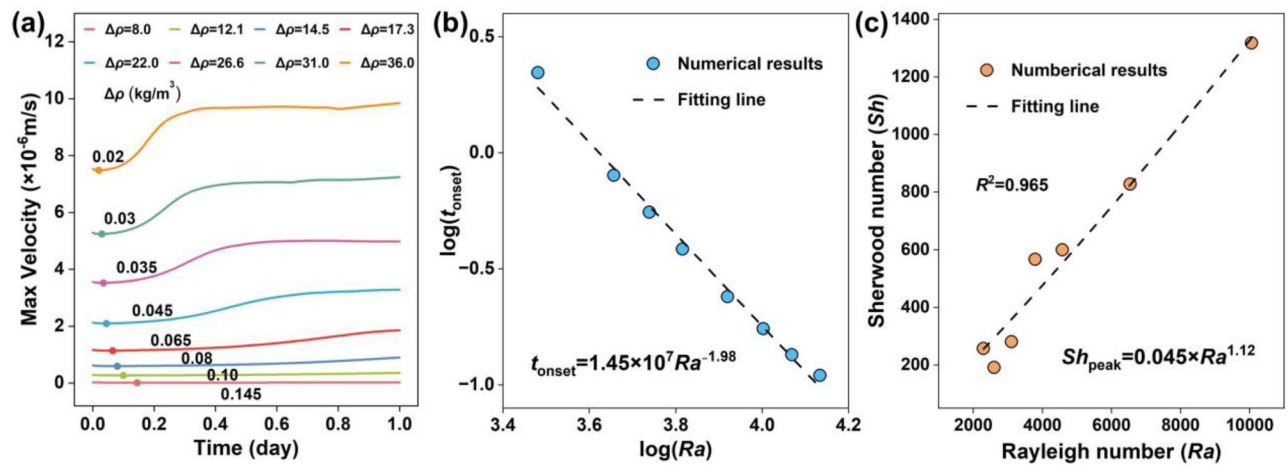


Fig. 13 Characterization of convective dissolution: (a) temporal evolution of maximum velocity for onset time determination; Scaling of (b) convection onset time and (c) peak Sherwood number as a function of Rayleigh number

Table 3 Rayleigh numbers and convection onset times for different $\Delta\rho$ cases

$\Delta\rho$ (kg/m ³)	$U_{\max}$ (m/s)	Onset time t_{onset} (day)	Ra
0.1	—	—	—
8.0	8.30×10^{-9}	0.145	3024
12.1	1.43×10^{-8}	0.100	4574
14.5	1.85×10^{-8}	0.080	5481
17.3	2.35×10^{-8}	0.065	6540
22.0	3.19×10^{-8}	0.045	8316
26.6	4.18×10^{-8}	0.035	10,055
31.0	5.37×10^{-8}	0.030	11,718
36	6.86×10^{-8}	0.020	13,608

Table 4 Correspondence between peak Sherwood number Sh_{peak} and the Rayleigh number Ra for different simulation cases

$\Delta\rho$ (kg/m ³)	Ra	Sh_{peak}	D (m ² /s)	Ra	Sh_{peak}
0.1	—	—	2.89×10^{-9}	3792	567
12.1	4574	600	3.53×10^{-9}	3107	281
17.3	6540	828	4.21×10^{-9}	2602	192
26.6	10,055	1317	4.78×10^{-9}	2295	257

exhibit an upward trend. This suggests that enhanced diffusion postpones the development of instabilities, thereby facilitating CO₂ accumulation and the formation of a thicker diffusive boundary layer prior to the initiation of convection.

Figure 12b shows that H_{eq} scales linearly with the diffusive characteristic length $\sqrt{Dt_{\text{onset}}}$,

$$H_{\text{eq}} = 2.52\sqrt{Dt_{\text{onset}}} + 7.80 \times 10^{-4} \quad (8)$$

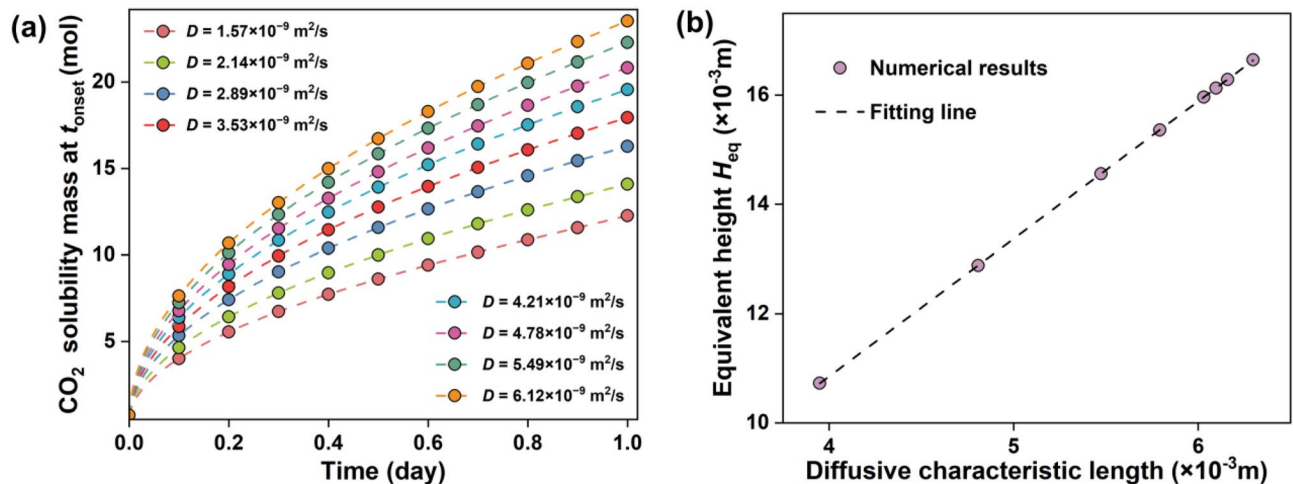
The growth of the pre instability boundary layer is primarily controlled by diffusion. From the plotted data, $\sqrt{Dt_{\text{onset}}}$ increases from 3.95×10^{-3} to 6.30×10^{-3} , while the corresponding equivalent thickness increases from

Table 5 Relevant parameters and scaling factors at the critical state of convective instability

D (m ² /s)	Dissolved CO ₂ mass at t_{onset} (mol)	H_{eq} (m)	$Ra = \frac{kg\Delta\rho H}{\mu D}$
1.57×10^{-9}	4.291	10.70×10^{-3}	0.00395
2.14×10^{-9}	5.123	12.90×10^{-3}	0.00481
2.89×10^{-9}	5.824	14.56×10^{-3}	0.00547
3.53×10^{-9}	6.145	15.36×10^{-3}	0.00579
4.21×10^{-9}	6.385	15.96×10^{-3}	0.00603
4.78×10^{-9}	6.451	16.13×10^{-3}	0.00610
5.49×10^{-9}	6.516	16.30×10^{-3}	0.00616
6.12×10^{-9}	6.659	16.60×10^{-3}	0.00630

about 10.70×10^{-3} to 16.60×10^{-3} . The growth of the dissolved boundary layer before instability remains diffusion controlled, and the convection starts only after a diffusion generated boundary layer has accumulated to a critical thickness.

The present simulations assume a homogeneous permeability field in order to isolate the effects of molecular diffusion coefficient and density contrast on convective onset and mass-transfer enhancement. In natural saline aquifers, permeability heterogeneity may modify fingering morphology, vertical migration pathways, and dissolution mass transfer. Upper high-permeability zones may promote preferential downward migration of CO₂-rich brine and enhance convective mass transfer, whereas upper low-permeability zones may restrict vertical exchange and delay fingering development. Layered sand-mud interbedded structures may further induce staged fingering and local channelization. Therefore, heterogeneity may introduce scatter or systematic deviations from the baseline relationships for t_{onset} and Sh_{peak} derived in this study. The proposed scaling laws should thus be interpreted as baseline relationships for homogeneous or

**Fig. 14** Determination of the critical equivalent mass thickness at convective onset. (a) Accumulated dissolved CO₂ mass. (b) Linear relationship between the critical equivalent height H_{eq} and the diffusive characteristic length $\sqrt{Dt_{\text{onset}}}$

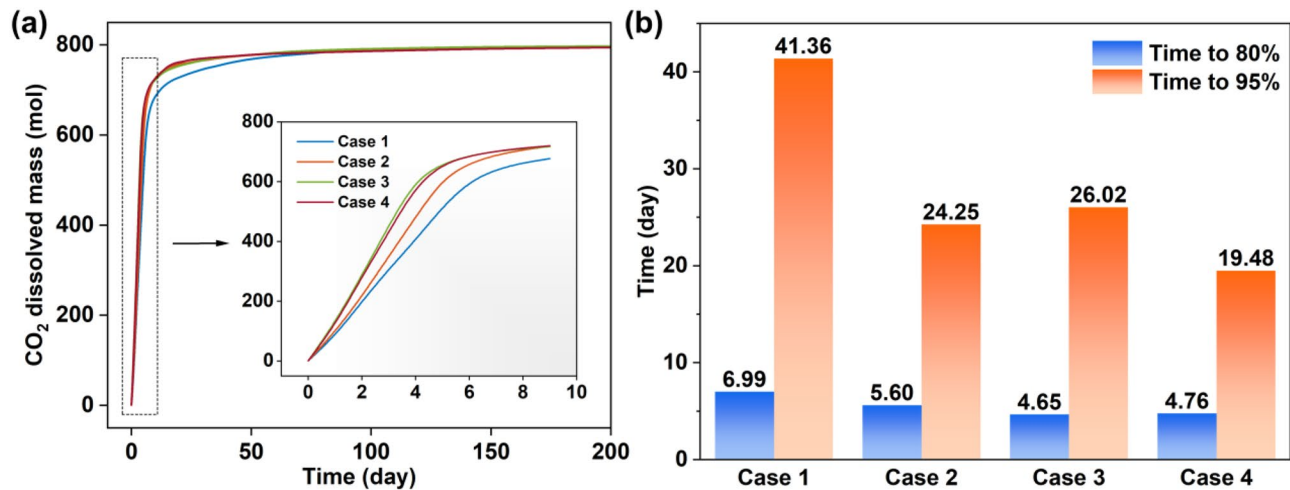


Fig. 15 Darcy scale prediction of dissolution performance for the representative South China Sea cases in the 2 m × 2 m model. (a) Temporal evolution of cumulative dissolved CO₂, with inset showing the early-stage. (b) Times required to reach 80% and 95% of the final dissolved CO₂ mass

effectively averaged porous media, while layered or stochastic permeability fields should be considered in future site-specific simulations.

It should be noted that the 2 m × 2 m numerical domain used in this study represents a local convective dissolution unit near the CO₂-brine interface rather than the full field-scale CO₂ plume. This domain size was selected to resolve the development of density-driven fingers and to isolate the effects of molecular diffusion coefficient and density contrast under controlled conditions. The resulting scaling relationships are therefore expressed in terms of dimensionless parameters, mainly Ra and Sh , and should be interpreted as baseline relationships for homogeneous local dissolution systems.

For field-scale offshore saline aquifers, the CO₂ plume may extend hundreds of meters laterally, and the vertical aquifer thickness may reach tens of meters. Under such conditions, domain height, lateral boundary conditions, plume geometry, and reservoir heterogeneity may affect finger spacing, finger coalescence, onset time, and mass-transfer rates. A larger vertical domain may allow longer finger growth and stronger late-stage interaction, whereas lateral boundaries and plume geometry may influence horizontal redistribution of CO₂-rich brine. Therefore, direct extrapolation of the present scaling laws to full-reservoir scale should be made with caution. Future work should couple the local scaling relationships with reservoir-scale models that include realistic aquifer thickness, lateral plume migration, and site-specific heterogeneity.

4.4 Engineering implications for offshore CO₂ storage in the South China Sea

The experimentally constrained results were further extended to representative CO₂ sequestration cases in

Table 6 Comparison of engineering performance indicators for the representative South China Sea cases

Performance Indicator	Case 1	Case 2	Case 3	Case 4
Cumulative dissolution at Day 5 (mol)	514.46	608.36	659.96	658.82
Final dissolved CO ₂ mass (mol)	800.41	800.85	801.45	801.16
Maximum daily dissolution (mol)	112.02	136.14	168.30	160.23
Time required for 80% saturation (day)	6.99	5.60	4.65	4.76
Time required for 95% saturation (day)	41.36	24.25	26.02	19.48

the South China Sea to evaluate the engineering significance of in situ fluid property variations. Four cases were defined to represent different temperature and pressure conditions, which affect CO₂ solubility, density contrast, diffusion coefficient, and viscosity. The corresponding parameters are listed in Table 2.

Figure 15(a) shows the temporal evolution of cumulative dissolved CO₂ for the four representative cases in the 2 m × 2 m Darcy scale model. All cases exhibit a similar overall trend, with rapid dissolution during the initial stage, followed by a gradual decline in dissolution rate and eventual approach to a stable limit. This behavior is consistent with the transport regimes identified in Sect. 4.2 and 4.3. By Day 5, the cumulative dissolved CO₂ reaches 514.46, 608.36, 659.96, and 658.82 mol for Cases 1 to 4 (Table 6), respectively. Relative to Case 1, the increase is about 18.2% for Case 2 and about 28.2% for Cases 3 and 4. These differences show that elevated temperature and pressure promote early-stage dissolution through the combined effects of larger molecular diffusivity and lower salinity corrected viscosity, which reduce transport resistance in the aqueous phase and enhance interfacial renewal.

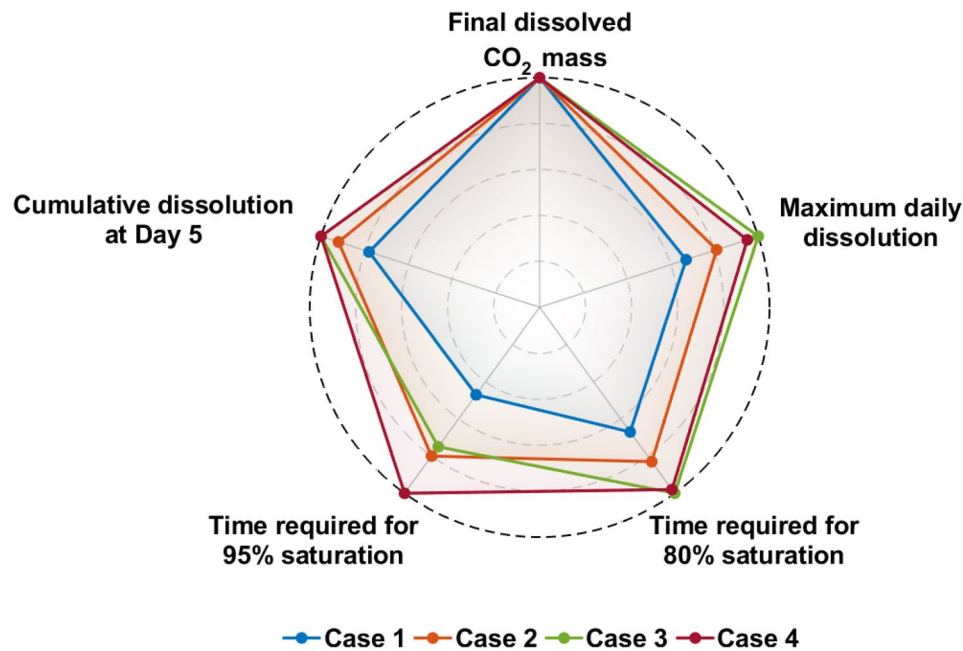


Fig. 16 Radar chart comparison of normalized engineering performance indicators for representative South China Sea storage cases

As the concentration gradient weakens and the system progressively approaches saturation, the dissolution rate decreases. Figure 15(b) compares the times required for each case to reach 80% and 95% of the final dissolved CO₂ mass. Case 3 reaches 80% saturation most rapidly, in 4.65 days, followed by Case 4 at 4.76 days, Case 2 at 5.60 days, and Case 1 at 6.99 days. This ranking shows that Case 3 has the strongest early-stage dissolution kinetics. However, its 95% saturation time is 26.02 days, longer than those of Case 2 and Case 4. By contrast, Case 4 reaches 95% saturation within 20 days, whereas Case 1 requires 41.36 days. This difference suggests that Case 3 is more favorable for rapid initial dissolution, while Case 4 maintains stronger mass transfer during the later stage. A similar distinction is reflected in the maximum daily dissolution. The highest value observed in Case 3. Despite these kinetic differences, the final dissolved amounts remain very close, about 800 mol. Therefore, under the present boundary conditions, fluid property variations have only a limited effect on the final dissolved CO₂ mass, but a much stronger influence on dissolution kinetics.

Table 6; Fig. 16 provide a more comprehensive comparison of engineering performance. Case 3 shows the highest maximum daily dissolution and the shortest time to 80% saturation, confirming its advantage in rapid early-stage convective dissolution. Case 4 shows the shortest time to 95% saturation while maintaining a high early stage dissolved amount and a large maximum daily dissolution, indicating a better balance between rapid onset and sustained transfer efficiency. Cases 1 and 2 perform

less favorably because weaker transport enhancement delays convective development and reduces dissolution efficiency over practical storage timescales. Overall, Case 4 provides the best integrated performance among the four representative scenarios.

From an engineering perspective, these results show that the practical performance of offshore CO₂ sequestration is governed less by the small difference in ultimate equilibrium capacity than by the kinetics of dissolution and plume stabilization over operationally relevant timescales. Faster conversion of mobile CO₂ into the dissolved phase shortens the persistence of buoyant free phase CO₂ and lowers the risk of remigration and leakage. Reservoir screening and injection design should therefore explicitly consider experimentally constrained in situ fluid properties, rather than relying only on generalized thermodynamic correlations. The present results provide a calibrated basis for identifying reservoir conditions that favor both rapid solubility trapping and secure long-term storage in the South China Sea.

5 Conclusion

This study developed an integrated experimental-numerical framework to investigate CO₂ convective dissolution in offshore saline aquifers relevant to the South China Sea. By combining measurements on actual reservoir-derived brines with Darcy-scale simulations, the study reduced the uncertainty associated with generalized fluid-property correlations and provided a physically constrained basis for evaluating density-driven convection.

The results show that the molecular diffusion coefficient of CO₂ in reservoir brines is strongly dependent on temperature, systematically reduced by salinity, and only weakly affected by pressure within the investigated range. In addition, CO₂ dissolution consistently increases brine density under all tested conditions, confirming the existence of the buoyancy contrast required to trigger convective mixing.

The numerical simulations further demonstrate that density-driven convection is governed by the competition between buoyancy destabilization and diffusive stabilization. A larger density contrast accelerates the onset of convection and strengthens fingering development, whereas a larger diffusion coefficient stabilizes the diffusive boundary layer and delays instability. The Rayleigh number is identified as the key parameter controlling both the transition to convection-dominated dissolution and the associated mass-transfer enhancement.

For representative South China Sea storage scenarios, reservoir conditions have a limited effect on final dissolved CO₂ mass, but a strong influence on dissolution kinetics and, therefore, on storage security over engineering timescales. The results indicate that reservoir screening and injection design should prioritize conditions that enhance early-stage convective dissolution rather than focusing solely on equilibrium storage capacity.

Overall, this study provides an experimentally grounded and application-oriented work for predicting and optimizing offshore CO₂ storage in saline aquifers. The proposed approach is directly relevant to reservoir screening, injection strategy design, and long-term storage security assessment in the South China Sea and other offshore basins.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1007/s43979-026-00178-7>.

Supplementary Material 1

Acknowledgements

We thank Shenzhen Guangming Science City Development and Construction Co., Ltd. and its Material Genome Big-Science Facilities Platform, for providing technical support and assistance in data collection and analysis.

Author contributions

Ying Teng and Pengfei Wang designed the study and wrote the manuscript. Qiang Han and Yiqi Chen performed simulations and data analysis. Jiaqi Wang contributed to visualization. Huiyu Sun and Songbai Han provided conceptual input and project coordination. Senyou An, Xuanlong Shan, Xi Chen, and Jinlong Zhu assisted in validation and methodology. Heping Xie supervised the project and acquired funding.

Funding

National Natural Science Foundation of China (52304098, 52474105), Youth S&T Talent Support Programme of Guangdong Provincial Association for Science and Technology (GDSTA), Natural Science Foundation of Guangdong Province (2025A1515010748), Shenzhen Science and Technology Program (JCYJ20220818095605012, SYSRD20250529113200001), Research Team

Cultivation Program of Shenzhen University (2023QNT004), Shenzhen University 2035 Initiative (2022B001).

Data availability

Available upon request.

Declarations

Competing interests

The authors declare that they have no competing interests.

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Received: 4 May 2026 / Revised: 30 June 2026 / Accepted: 22 July 2026

Published online: 29 July 2026

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